

Machine Learning Pipeline :

Data Collection .1

- تسجيل أوامر صوتية حقيقية
- تخزينها بصيغة WAV

Preprocessing .2

- Normalization
- kHz Sampling Rate = 16

Feature Extraction .3

- MFCC
- Delta-Delta & Delta
- mean, std,)Statistical Features
- (min, max

Feature Scaling .4

- StandardScaler

Dimensionality Reduction .5

- PCA (Visualization)

Model Training .6

- Random Forest Classifier

Evaluation .7

- Accuracy
- Confusion Matrix
- Classification Report

Deployment .8

- Real-time prediction
- Arduino control via Serial



Technologies Used

البرمجيات

- Python
- Librosa
- Scikit-learn
- Sounddevice
- 3Pyttax
- Joblib

الذكاء الاصطناعي

- Speech Recognition
- Feature Engineering
- Supervised Learning
- Random)Ensemble Learning
- (Forest

العتاد

- Arduino
- Microphone Module
- (Gesture / Sensors (Parking
- Serial Communication

